

From: Treatment Perspectives™ on behalf of Merck [education@treatmentperspectives.com]

To: [HCP email]

Option 1

Subject Line: Review clinical data for a T2DM treatment option

Preheader: See the tolerability data from a T2DM study

Option 2

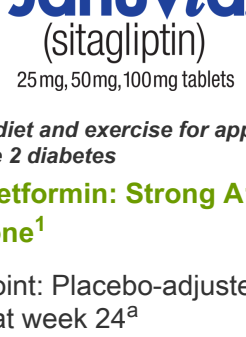
Subject Line: Clinical trial results in adults with T2DM inadequately controlled on metformin

Preheader: See the tolerability data from a T2DM study

Option 3

Subject Line: Learn about clinical study data for this T2DM treatment option

Preheader: See the tolerability data from a T2DM study

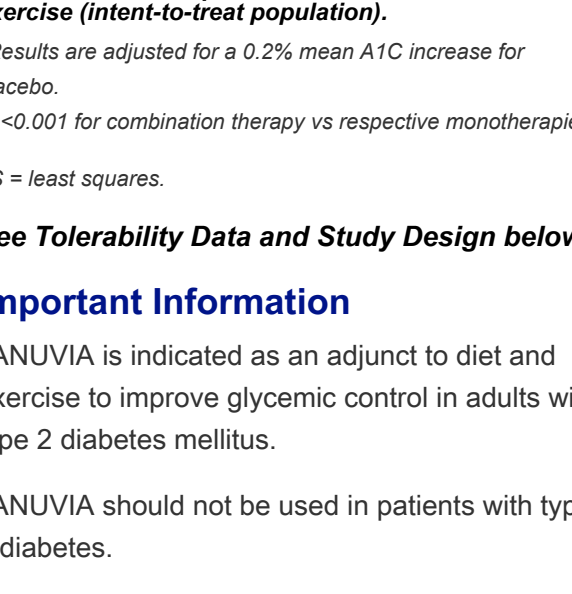


25 mg, 50 mg, 100 mg tablets

As an adjunct to diet and exercise for appropriate adult patients with type 2 diabetes

JANUVIA + metformin: Strong A1C lowering vs metformin alone¹

Primary end point: Placebo-adjusted change from baseline A1C at week 24^a



Week 24 results for patients uncontrolled on diet and exercise (intent-to-treat population).

^aResults are adjusted for a 0.2% mean A1C increase for placebo.

^bP<0.001 for combination therapy vs respective monotherapies.

LS = least squares.

See Tolerability Data and Study Design below.

Important Information

JANUVIA is indicated as an adjunct to diet and exercise to improve glycemic control in adults with type 2 diabetes mellitus.

JANUVIA should not be used in patients with type 1 diabetes.

JANUVIA has not been studied in patients with a history of pancreatitis. It is unknown whether patients with a history of pancreatitis are at increased risk of developing pancreatitis while taking JANUVIA.

Selected Important Risk Information

JANUVIA is contraindicated in patients with a history of a serious hypersensitivity reaction to sitagliptin, such as anaphylaxis or angioedema.

There have been postmarketing reports of acute pancreatitis, including fatal and nonfatal hemorrhagic or necrotizing pancreatitis, in patients taking JANUVIA. After initiating JANUVIA, observe patients carefully for signs and symptoms of pancreatitis. If pancreatitis is suspected, promptly discontinue JANUVIA and initiate appropriate management. It is unknown whether patients with a history of pancreatitis are at increased risk of developing pancreatitis while taking JANUVIA.

An association between dipeptidyl peptidase-4 (DPP-4) inhibitor treatment and heart failure has been observed in cardiovascular outcomes trials for two other members of the DPP-4 inhibitor class. These trials evaluated patients with type 2 diabetes mellitus and atherosclerotic cardiovascular disease. Consider the risks and benefits of JANUVIA prior to initiating treatment in patients at risk for heart failure, such as those with a prior history of heart failure and a history of renal impairment, and observe these patients for signs and symptoms of heart failure during therapy.

Advise patients of the characteristic symptoms of heart failure and to immediately report such symptoms. If heart failure develops, evaluate and manage according to current standards of care and consider discontinuation of JANUVIA.

Selected Important Risk Information continued below.

Established Tolerability Profile

Similar rates of hypoglycemia with JANUVIA + metformin vs metformin alone¹

Percentage of patients with ≥ 1 episode of hypoglycemia over 24 weeks (APaT population)

Placebo (n=176)	0.6%
JANUVIA 100 mg daily (n=179)	0.6%
Metformin 1000 mg daily (n=182)	0.5%
Metformin 2000 mg daily (n=182)	1.1%
JANUVIA + metformin 100/1000 mg daily (n=190)	1.1%
JANUVIA + metformin 100/2000 mg daily (n=182)	2.2%

APaT, all-patients-as-treated.

When added to a sulfonylurea (glimepiride) or insulin, patients treated with JANUVIA experienced increased incidence of hypoglycemia.

12.2% for patients treated with JANUVIA + glimepiride $\pm$ metformin vs 1.8% for placebo + glimepiride $\pm$ metformin; 15.5% for patients treated with JANUVIA + insulin $\pm$ metformin vs 7.8% for placebo + insulin $\pm$ metformin.

A lower dose of sulfonylurea or insulin may be required to reduce the risk of hypoglycemia.

See Tolerability Data and Study Design below.

Selected Important Risk Information (cont.)

There have been postmarketing reports of worsening renal function, including acute renal failure, sometimes requiring dialysis. A subset of these reports involved patients with renal impairment, some of whom were prescribed inappropriate doses of sitagliptin.

Assessment of renal function is recommended prior to initiating JANUVIA and periodically thereafter. A dosage adjustment is recommended in patients with moderate or severe renal impairment and in patients with end-stage renal disease requiring hemodialysis or peritoneal dialysis.

Selected Important Risk information continued below.

Similar weight change with JANUVIA + metformin vs metformin alone¹

Weight loss for JANUVIA + metformin vs metformin alone

	LS mean change from baseline at 24 weeks (APaT population)
JANUVIA + metformin	-1.3 lb to -2.9 lb
Metformin	-1.3 lb to -2.9 lb

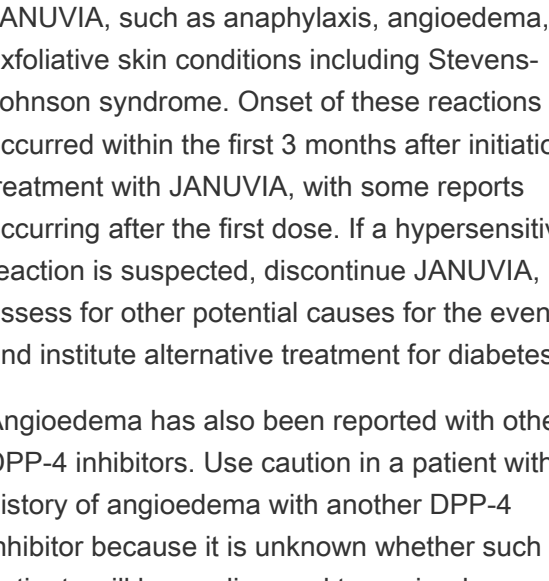
LS = least squares; APaT = all-patients-as-treated.

When added to a sulfonylurea (glimepiride) or insulin, patients treated with JANUVIA experienced a mean increase in body weight.

+1.8 lb for patients treated with JANUVIA + glimepiride $\pm$ metformin vs -0.9 lb for placebo + glimepiride $\pm$ metformin; +0.2 lb for patients treated with JANUVIA + insulin $\pm$ metformin vs +0.2 lb for placebo + insulin $\pm$ metformin.

Similar rate of GI events with JANUVIA + metformin vs metformin alone¹

Incidence of all GI adverse events for JANUVIA + metformin vs metformin alone over 24 weeks^{a,b}



^aIncidence of all GI adverse events: placebo (n=176), 10.8%; JANUVIA 100 mg daily (n=179), 15.1%.

^bPercentage of patients (APaT population).

GI = gastrointestinal; APaT = all-patients-as-treated.

The incidence of selected gastrointestinal adverse reactions in patients treated with JANUVIA and metformin was similar to those of placebo and metformin: nausea (1.3%, 0.8%), vomiting (1.1%, 0.8%), abdominal pain (2.2%, 3.8%) and diarrhea (2.4%, 2.5%).

See Study Design below.

Check Formulary Coverage Here

[Click Here](#)

Selected Important Risk Information (cont.)

When JANUVIA was used in combination with insulin or insulin secretagogues (eg, sulfonylurea), medications known to cause hypoglycemia, the incidence of hypoglycemia was increased over that of placebo. Therefore, a lower dose of sulfonylurea or insulin may be required to reduce the risk of hypoglycemia.

The incidence (and rate) of hypoglycemia based on all reports of symptomatic hypoglycemia were: 12.2% (0.59 episodes/patient-year) for JANUVIA 100 mg in combination with glimepiride (with or without metformin), 1.8% (0.24 episodes/patient-year) for placebo in combination with glimepiride (with or without metformin), 15.5% (1.06 episodes/patient-year) for JANUVIA 100 mg in combination with insulin (with or without metformin), and 7.8% (0.51 episodes/patient-year) for placebo in combination with insulin (with or without metformin).

There have been postmarketing reports of serious hypersensitivity reactions in patients treated with JANUVIA, such as anaphylaxis, angioedema, and exfoliative skin conditions including Stevens-Johnson syndrome. Onset of these reactions occurred within the first 3 months after initiation of treatment with JANUVIA, with some reports occurring after the first dose. If a hypersensitivity reaction is suspected, discontinue JANUVIA, assess for other potential causes for the event, and institute alternative treatment for diabetes.

Angioedema has also been reported with other DPP-4 inhibitors. Use caution in a patient with a history of angioedema with another DPP-4 inhibitor because it is unknown whether such patients will be predisposed to angioedema with JANUVIA.

There have been postmarketing reports of severe and disabling arthralgia in patients taking DPP-4 inhibitors. The time to onset of symptoms following initiation of drug therapy varied from 1 day to years. Patients experienced relief of symptoms upon discontinuation of the medication. A subset of patients experienced a recurrence of symptoms when restarting the same drug or a different DPP-4 inhibitor. Consider DPP-4 inhibitors as a possible cause for severe joint pain and discontinue drug if appropriate.

Postmarketing cases of bullous pemphigoid requiring hospitalization have been reported with DPP-4 inhibitor use. In reported cases, patients typically recovered with topical or systemic immunosuppressive treatment and discontinuation of the DPP-4 inhibitor. Tell patients to report development of blisters or erosions while receiving JANUVIA. If bullous pemphigoid is suspected, JANUVIA should be discontinued and referral to a dermatologist should be considered for diagnosis and appropriate treatment.

In clinical studies, the adverse reactions reported, regardless of investigator assessment of causality, in $\geq 5\%$ of patients treated with JANUVIA as monotherapy and in combination therapy and more commonly than in patients treated with placebo, were upper respiratory tract infection, nasopharyngitis, and headache.

Study Design

Study evaluating the efficacy and safety of initial combination therapy with sitagliptin and metformin vs metformin or sitagliptin alone: A total of 1,091 patients with type 2 diabetes and inadequate glycemic control on diet and exercise participated in a randomized, double-blind, placebo-controlled factorial study designed to assess the efficacy and safety of sitagliptin + metformin compared with respective monotherapies. Patients were randomized into 1 of 6 treatment groups: sitagliptin + metformin 50/500 mg bid (n=190), sitagliptin + metformin 50/1000 mg bid (n=182), metformin 500 mg bid (n=182), metformin 1000 mg bid (n=182), sitagliptin 100 mg once daily (n=179), or placebo (n=176). The primary end point was measured after 24 weeks of treatment.¹

bid = twice daily.

Before prescribing JANUVIA® (sitagliptin) please read the accompanying [Prescribing Information](#). The [Medication Guide](#) also is available.

Reference:

1. Goldstein BJ, Feinglos MN, Lunceford JK, et al; for Sitagliptin 036 Study Group. Effect of initial combination therapy with sitagliptin, a dipeptidyl peptidase-4 inhibitor, and metformin on glyemic control in patients with type 2 diabetes. *Diabetes Care*. 2007;30(8):1979-1987.

For prescribers: please [click here](#) for state-required price disclosures.

Note: This message was distributed from an email account used for sending messages only. Please do not reply to this message.

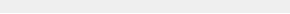
You may have elected to receive email communications from Merck Sharp & Dohme LLC. You can unsubscribe by clicking [here](#) anytime. If you have trouble with this link, please send an email to merck_privacy_office@merck.com with a subject line of Email Opt-Out Request - XXXXX

Copyright © 2023 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.

To learn more about Merck's commitment to privacy and your rights, please visit [merck.com/privacy](https://www.merck.com/privacy).

Merck Privacy Office, 351 N Sumneytown Pike, UG4B-24, North Wales, PA 19454, USA

US-DIA-02648 08/23



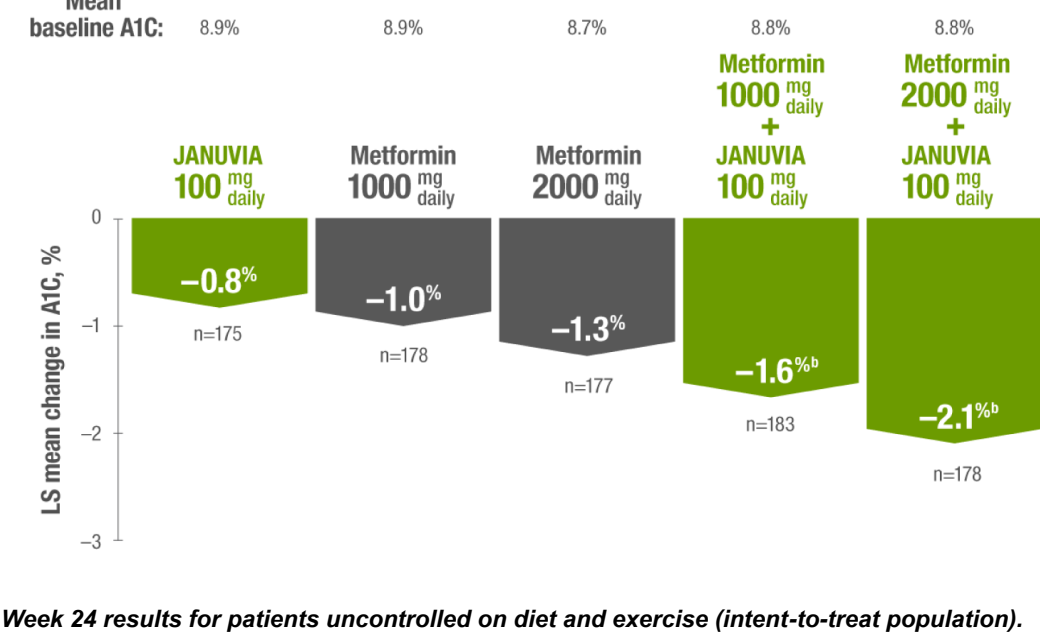


25mg, 50mg, 100mg tablets

As an adjunct to diet and exercise for appropriate adult patients with type 2 diabetes

JANUVIA + metformin: Strong A1C lowering vs metformin alone¹

Primary end point: Placebo-adjusted change from baseline A1C at week 24^a



Week 24 results for patients uncontrolled on diet and exercise (intent-to-treat population).

^aResults are adjusted for a 0.2% mean A1C increase for placebo.

^bP<0.001 for combination therapy vs respective monotherapies.

LS = least squares.

See **Tolerability Data and Study Design** below.

Important Information

JANUVIA is indicated as an adjunct to diet and exercise to improve glycemic control in adults with type 2 diabetes mellitus.

JANUVIA should not be used in patients with type 1 diabetes.

JANUVIA has not been studied in patients with a history of pancreatitis. It is unknown whether patients with a history of pancreatitis are at increased risk of developing pancreatitis while taking JANUVIA.

Selected Important Risk Information

JANUVIA is contraindicated in patients with a history of a serious hypersensitivity reaction to sitagliptin, such as anaphylaxis or angioedema.

There have been postmarketing reports of acute pancreatitis, including fatal and nonfatal hemorrhagic or necrotizing pancreatitis, in patients taking JANUVIA. After initiating JANUVIA, observe patients carefully for signs and symptoms of pancreatitis. If pancreatitis is suspected, promptly discontinue JANUVIA and initiate appropriate management. It is unknown whether patients with a history of pancreatitis are at increased risk of developing pancreatitis while taking JANUVIA.

An association between dipeptidyl peptidase-4 (DPP-4) inhibitor treatment and heart failure has been observed in cardiovascular outcomes trials for two other members of the DPP-4 inhibitor class. These trials evaluated patients with type 2 diabetes mellitus and atherosclerotic cardiovascular disease. Consider the risks and benefits of JANUVIA prior to initiating treatment in patients at risk for heart failure, such as those with a prior history of heart failure and a history of renal impairment, and observe these patients for signs and symptoms of heart failure during therapy. Advise patients of the characteristic symptoms of heart failure and to immediately report such symptoms. If heart failure develops, evaluate and manage according to current standards of care and consider discontinuation of JANUVIA.

Selected Important Risk Information continued below.

Established Tolerability Profile

Similar rates of hypoglycemia with JANUVIA + metformin vs metformin alone¹

Percentage of patients with ≥ 1 episode of hypoglycemia over 24 weeks (APaT population)

Placebo (n=176)	0.6%
JANUVIA 100 mg daily (n=179)	0.6%
Metformin 1000 mg daily (n=182)	0.5%
Metformin 2000 mg daily (n=182)	1.1%
JANUVIA + metformin 100/1000 mg daily (n=190)	1.1%
JANUVIA + metformin 100/2000 mg daily (n=182)	2.2%

APaT, all-patients-as-treated.

When added to a sulfonylurea (glimepiride) or insulin, patients treated with JANUVIA experienced increased incidence of hypoglycemia.

12.2% for patients treated with JANUVIA + glimepiride $\pm$ metformin vs 1.8% for placebo + glimepiride $\pm$ metformin; 15.5% for patients treated with JANUVIA + insulin $\pm$ metformin vs 7.8% for placebo + insulin $\pm$ metformin.

A lower dose of sulfonylurea or insulin may be required to reduce the risk of hypoglycemia.

See **Tolerability Data and Study Design** below.

Selected Important Risk Information (cont.)

There have been postmarketing reports of worsening renal function, including acute renal failure, sometimes requiring dialysis. A subset of these reports involved patients with renal impairment, some of whom were prescribed inappropriate doses of sitagliptin.

Assessment of renal function is recommended prior to initiating JANUVIA and periodically thereafter. A dosage adjustment is recommended in patients with moderate or severe renal impairment and in patients with end-stage renal disease requiring hemodialysis or peritoneal dialysis.

Selected Important Risk information continued below.

Similar weight change with JANUVIA + metformin vs metformin alone¹

Weight loss for JANUVIA + metformin vs metformin alone

	LS mean change from baseline at 24 weeks (APaT population)
JANUVIA + metformin	-1.3 lb to -2.9 lb
Metformin	-1.3 lb to -2.9 lb

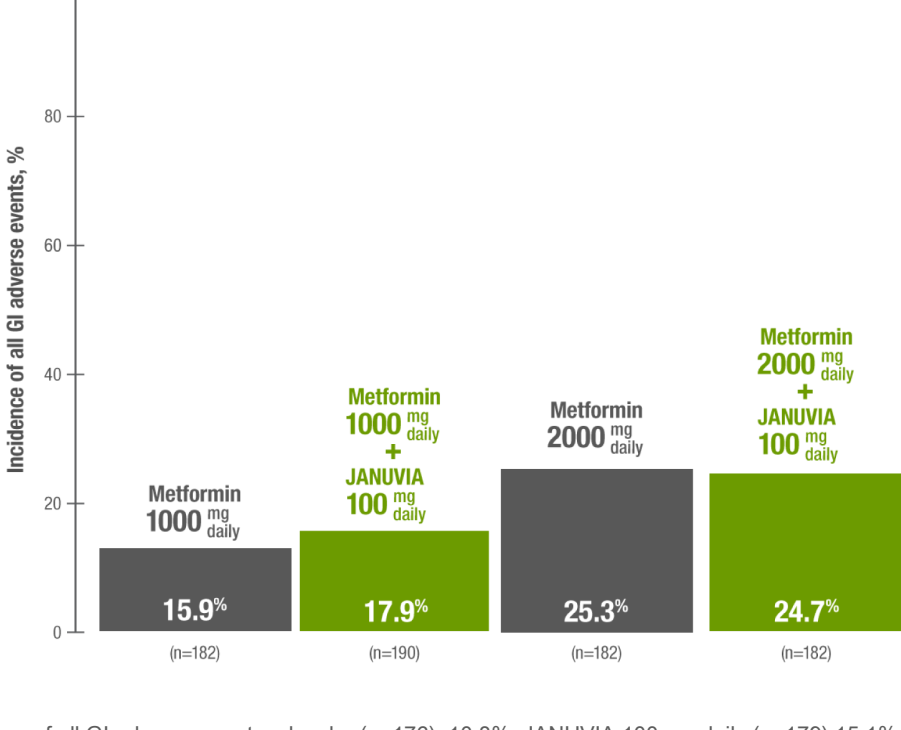
LS = least squares; APaT = all-patients-as-treated.

When added to a sulfonylurea (glimepiride) or insulin, patients treated with JANUVIA experienced a mean increase in body weight.

+1.8 lb for patients treated with JANUVIA + glimepiride $\pm$ metformin vs -0.9 lb for placebo + glimepiride $\pm$ metformin; +0.2 lb for patients treated with JANUVIA + insulin $\pm$ metformin vs +0.2 lb for placebo + insulin $\pm$ metformin.

Similar rate of GI events with JANUVIA + metformin vs metformin alone¹

Incidence of all GI adverse events for JANUVIA + metformin vs metformin alone over 24 weeks^{a,b}



^aIncidence of all GI adverse events: placebo (n=176), 10.8%; JANUVIA 100 mg daily (n=179), 15.1%.

^bPercentage of patients (APaT population).

GI = gastrointestinal; APaT = all-patients-as-treated.

The incidence of selected gastrointestinal adverse reactions in patients treated with JANUVIA and metformin was similar to those of placebo and metformin: nausea (1.3%, 0.8%), vomiting (1.1%, 0.8%), abdominal pain (2.2%, 3.8%) and diarrhea (2.4%, 2.5%).

See **Study Design** below.

Check Formulary Coverage Here

[Click Here](#)

Selected Important Risk Information (cont.)

When JANUVIA was used in combination with insulin or insulin secretagogues (eg, sulfonylurea), medications known to cause hypoglycemia, the incidence of hypoglycemia was increased over that of placebo. Therefore, a lower dose of sulfonylurea or insulin may be required to reduce the risk of hypoglycemia.

The incidence (and rate) of hypoglycemia based on all reports of symptomatic hypoglycemia were: 12.2% (0.59 episodes/patient-year) for JANUVIA 100 mg in combination with glimepiride (with or without metformin), 1.8% (0.24 episodes/patient-year) for placebo in combination with glimepiride (with or without metformin), 15.5% (1.06 episodes/patient-year) for JANUVIA 100 mg in combination with insulin (with or without metformin), and 7.8% (0.51 episodes/patient-year) for placebo in combination with insulin (with or without metformin).

There have been postmarketing reports of serious hypersensitivity reactions in patients treated with JANUVIA, such as anaphylaxis, angioedema, and exfoliative skin conditions including Stevens-Johnson syndrome. Onset of these reactions occurred within the first 3 months after initiation of treatment with JANUVIA, with some reports occurring after the first dose. If a hypersensitivity reaction is suspected, discontinue JANUVIA, assess for other potential causes for the event, and institute alternative treatment for diabetes.

Angioedema has also been reported with other DPP-4 inhibitors. Use caution in a patient with a history of angioedema with another DPP-4 inhibitor because it is unknown whether such patients will be predisposed to angioedema with JANUVIA.

There have been postmarketing reports of severe and disabling arthralgia in patients taking DPP-4 inhibitors. The time to onset of symptoms following initiation of drug therapy varied from 1 day to years. Patients experienced relief of symptoms upon discontinuation of the medication. A subset of patients experienced a recurrence of symptoms when restarting the same drug or a different DPP-4 inhibitor. Consider DPP-4 inhibitors as a possible cause for severe joint pain and discontinue drug if appropriate.

Postmarketing cases of bullous pemphigoid requiring hospitalization have been reported with DPP-4 inhibitors. In reported cases, patients typically recovered with topical or systemic immunosuppressive treatment and discontinuation of the DPP-4 inhibitor. Tell patients to report development of blisters or erosions while receiving JANUVIA. If bullous pemphigoid is suspected, JANUVIA should be discontinued and referral to a dermatologist should be considered for diagnosis and appropriate treatment.

In clinical studies, the adverse reactions reported, regardless of investigator assessment of causality, in $\geq 5\%$ of patients treated with JANUVIA as monotherapy and in combination therapy and more commonly than in patients treated with placebo, were upper respiratory tract infection, nasopharyngitis, and headache.

Study Design

Study evaluating the efficacy and safety of initial combination therapy with sitagliptin and metformin vs metformin or sitagliptin alone: A total of 1,091 patients with type 2 diabetes and inadequate glycemic control on diet and exercise participated in a randomized, double-blind, placebo-controlled factorial study designed to assess the efficacy and safety of sitagliptin + metformin compared with respective monotherapies. Patients were randomized into 1 of 6 treatment groups: sitagliptin + metformin 50/500 mg bid (n=190), sitagliptin + metformin 50/1000 mg bid (n=182), metformin 500 mg bid (n=182), metformin 1000 mg bid (n=182), sitagliptin 100 mg once daily (n=179), or placebo (n=176). The primary end point was measured after 24 weeks of treatment.¹

bid = twice daily.

Before prescribing JANUVIA® (sitagliptin) please read the accompanying Prescribing Information. The Medication Guide also is available.

Reference:

1. Goldstein BJ, Feinglos MN, Lunceford JK, et al, for Sitagliptin 036 Study Group. Effect of initial combination therapy with sitagliptin, a dipeptidyl peptidase-4 inhibitor, and metformin on glycemic control in patients with type 2 diabetes. *Diabetes Care*. 2007;30(8):1979-1987.

For prescribers: please [click here](#) for state-required price disclosures.

Note: This message was distributed from an email account used for sending messages only. Please do not reply to this message.

You may have elected to receive email communications from Merck Sharp & Dohme LLC. You can unsubscribe by clicking [here](#) anytime. If you have trouble with this link, please send an email to merck_privacy_office@merck.com with a subject line of Email Opt-Out Request - XXXXX

Copyright © 2023 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.

To learn more about Merck's commitment to privacy and your rights, please visit merck.com/privacy.

Merck Privacy Office, 351 N Summeytown Pike, UG4B-24, North Wales, PA 19454, USA

US-DIA-02648 08/23

